Student Attendance Analytics System

Project Plan v2.0: Problem, Solution, Roles, Screens and Database Design

Core idea: Problem → Data → Analytics → Insight → Decision → Action → Measured result

> **In one sentence:** This is not just a system that records who is present. It turns attendance data into useful knowledge, so a high school can spot problems early, decide which students need help first, act, and check whether the action worked.

Version	2.0 (replaces v1.0)
School type	High school (Grades 10–12)
Platforms	Android app (teachers) and website (admins and teachers)
User roles	2: Admin and Teacher
Backend	Laravel REST API + PostgreSQL (one shared backend and database)

Contents

1. The big idea: analytics, not just recording
2. The problems the system solves
3. The five layers of analytics
4. What the system analyzes and shows
5. Android app and website
6. Problem statement, solution and benefits
7. How teachers record attendance
8. User roles and permissions (2 roles)
9. Early warning, interventions and parent messages
10. Screens
11. Database design (PostgreSQL)
12. Analytics rules and formulas
13. Scope, timeline and development order
14. Testing, privacy and limitations
15. Appendix: what changed from v1.0

1. The big idea: analytics, not just recording

Most attendance systems only save data and show a list of who was present and who was absent. In this project, attendance is only the **source of data**. The real value is the **analytics** on that data. The system should help answer:

- What is happening?
- Where and when is it happening (class, subject, day, period)?
- Which students have attendance problems, and how serious are they?
- Is it getting better or worse?
- Who should the school help first?
- Did the school's action help?

> That is what makes this an **Analytics System** and not only an *Attendance Recording System*.

2. The problems the system solves

Problem 1: The school knows a student is absent, but not the pattern

Example: in one month Dara was Present 18 times and Absent 5 times. "Dara is absent a lot" is not deep enough. The system should find a **pattern**:

- Dara is rarely absent on Monday and Friday.
- Dara is often absent on Wednesday, mostly in Period 1.
- Most of those absences are in Mathematics.
- Absences have increased over the last 4 weeks.

> **Analytics result:** "Dara has a repeated absence pattern in Wednesday Period 1 Mathematics, and attendance has declined over the last four weeks."

Problem 2: The attendance rate alone can hide the truth

Attendance Rate = Present ÷ all submitted records × 100%. Two students can both have 90% and be very different:

Week	Student A	Student B
Week 1	100%	90%
Week 2	100%	90%
Week 3	100%	90%
Week 4	80%	90%
Week 5	70%	90%
Average	**90%**	**90%**

Student A needs more attention because the trend is falling. The system always shows the rate **together with its trend**, for example: **87% attendance, down 8 points from the previous 4 weeks, declining.**

Problem 3: Schools find problems too late

A student's attendance may go 95% → 90% → 82% → 68% over four months. If the school reacts only in month 4, it is reactive. The system should be proactive:

Data → Detect pattern → Identify risk → Alert → Intervention → Measure result

Warning signs the system detects:

- Attendance falling for 3 weeks in a row
- Absent 3 or more school days in a row
- Repeated absence in the same subject or period
- Absences concentrated on certain days
- A sudden drop compared with the student's own past

From these signs the system gives each student a **risk level** (exact rules in Section 12):

Student	Attendance	Trend (4 weeks)	Longest run of absences	Pattern	Risk level
Dara	88%	Down 6 points	2 days	Wednesday P1	Medium
Sokha	78%	Down 12 points	3 days	Increasing	High
Lina	96%	Stable	1 day	Normal	Low
Vuthy	68%	Down 15 points	5 days	Increasing	Critical

Problem 4: Is the problem about one student, or about the timetable?

If Class 10A has low attendance, a school may think the students lack discipline. Analytics may show: Mathematics 95%, Khmer 94%, English 92%, **Physics 76%**. Digging further (period, day, room) may show that **Physics, Period 1, Monday** is the issue. Possible causes include the schedule, transport, room location, subject difficulty or a timetable conflict.

> **Important:** analytics must never blame a teacher. It only shows patterns that the school may want to look into.

Problem 5: Schools don't know when and where problems happen

Time-based analytics finds **hotspots**. Example: by day, Wednesday is lowest (86%). Within Wednesday, Period 1 is 78%, Period 2 is 91% and Period 3 is 93%. So **Wednesday Period 1 is a hotspot**, which is hard to see in a plain Excel table.

Problem 6: Not all absences mean the same thing

Two students may both have 8 absences. Student A: 6 sick, 2 family. Student B: 8 with no reason given. These are very different situations. The attendance status stays **Present or Absent only**, but an absent record can have an **optional reason**: Sick, Family matter, School activity, Other, or Not given (default). The system shows the **absence reason breakdown** for each student and class.

Problem 7: Absences in a row matter more than totals

Student A is absent, present, absent, present. Student B is absent 4 days in a row. The totals can be equal, but Student B is more worrying. The system creates a **consecutive absence alert** ("absent 3 school days in a row") for the student's teachers and the Admin.

Problem 8: Who should be helped first?

If a school has 1,500 students and 300 are below 90%, a plain list does not help. The system gives a **priority list** (Critical, High, Medium, Low) based on attendance level, trend, absences in a row and unexplained absences.

Problem 9: Did the school's action work?

Suppose the school contacts a student's parents. The system compares attendance in the 2 weeks **before** and the 2 weeks **after**:

- Before 68%, after 87%: **improved** (+19 points).
- Before 78%, after 61%: **declined**, so the school may need a different approach.

This creates a feedback loop: **Detect → Intervene → Measure → Evaluate → Improve.**

3. The five layers of analytics

Layer	Question	Examples in this system
1. Descriptive	What happened?	Attendance rate and absent count by student, class, grade, subject and month
2. Diagnostic	Where and when is it happening?	Lowest subject, lowest day, lowest period, day × period hotspots, absence reasons, trend direction
3. Early warning	Who is at risk if this continues?	Risk level (Critical/High/Medium/Low), consecutive absence alerts, "falling 3 weeks in a row"
4. Suggested action	What could the school consider?	For a High-risk student: review the reason, teacher follow-up, contact the parent, monitor for 2 weeks
5. Intervention result	Did our action help?	Before 70%, after 88%: improved by 18 points

> **Two cautions.** (1) Layer 3 uses the words **At-Risk** and **Early Warning**. The system does not claim to predict dropouts; that would need a large historical dataset and a tested model. (2) Layer 4 **supports decisions; people decide**. Suggestions are never automatic actions.

4. What the system analyzes and shows

Analytics groups

Group	What it includes
Student	Rate, trend, absences in a row, absence reasons, rate by subject, day and period pattern, risk level, interventions
Class	Rate and trend, number of at-risk students, sessions not yet recorded, comparison between classes
Grade	Grade 10 vs 11 vs 12: rate, trend, risk distribution
Subject	Rate by subject, subject × period patterns
Time	Day of week, period, week, month, term; day × period hotspot grid

Group	What it includes
Intervention	Number and type of interventions, before vs after rate, share improved, cases needing follow-up

Every number needs a meaning

Level	What the dashboard says
Weak	Attendance = 84%
Better	Attendance = 84%, down 7 points from the previous 4 weeks
Even better	Attendance = 84%, down 7 points, with 62% of absences in Period 1
Best (target)	Attendance is declining, especially in Period 1. 12 students moved into High risk and should be reviewed.

Admin dashboard (website home page)

Indicator	Example value
Attendance this month	91.4% (down 2.8 points)
Students at High or Critical risk	37
Students absent 3+ days in a row	12
Sessions not yet recorded today	4 of 48
Classes with declining attendance	5
Open interventions needing follow-up	9

Below these, the dashboard shows the monthly trend, the rate by grade (for example Grade 10: 86%, Grade 11: 93%, Grade 12: 94%) and short written insights such as *"Grade 10 shows the largest decline this term."*

Drill-down: School → Grade → Class → Student

The Admin clicks **Grade 10** and sees each class (10A 94%, 10B 91%, **10C 84%**, 10D 93%). Clicking **10C** shows: rate 84%, down 9 points, 8 High-risk students, lowest day Wednesday, lowest period 1, lowest subject Physics. Clicking a student opens the **Student Profile**:

Field	Example (student Dara)
Attendance and trend	78%, down 14 points
Longest run of absences (current)	3 school days
Most absences	Wednesday, Period 1, Physics
Absence reasons	Sick 2 · Family 1 · Not given 4
Risk level	High
Latest intervention	Parent contacted by Telegram, then phone call
Before → after	72% → 78% (+6 points)
Status	Improving, continue monitoring

5. Android app and website

The two platforms have **different jobs**, so they do not have the same interface. Both use the same backend and the same data.

	Android app	Website
Used by	Teachers	Admins (whole school) and Teachers (own classes only)
Main job	Collect data in the classroom and act at once	Analytics, monitoring, decisions and school management
Main features	Login; today's sessions; mark Present/Absent; optional absence reason; review and submit; notify parents of absent students; class history; alerts for own students; record a follow-up	Admin: school dashboard, drill-down, risk list, hotspots, interventions, corrections, reports and export, school data management, audit log. Teacher: own dashboard, take attendance on a computer, history, own classes

> **Android = data collection + immediate action. Website = analytics + monitoring + decision support.** Teachers can also use the website on a computer, but they only see their own classes.

6. Problem statement, solution and benefits

Core problem statement

> High schools collect a lot of attendance data, but they usually lack a way to turn that data into clear information about attendance patterns, declining trends, at-risk students and the effect of their actions. As a result, problems are found late, and decisions about which students to help are not always based on enough evidence.

"Paper records are slow" is **not** the main problem. Fast recording is only the first step, because without good data there is nothing to analyze.

Proposed solution

> The Student Attendance Analytics System collects attendance quickly on an Android app and the website, then analyzes it to find patterns, trends, risk levels and early-warning signals. Dashboards, alerts and reports help teachers and admins decide which students to follow up first, record what they did, and measure whether attendance improved afterwards.

The system pipeline

Schedule → Attendance session → Attendance records → Analytics → Patterns and trends → Risk level → Alert → Intervention → Measure result

Benefits

Benefit	Meaning
Early detection	Find students who are struggling before their attendance drops sharply
Better decisions	Admins use data instead of feelings or guesses
Hidden patterns	Show day, period, subject, class and trend patterns that a plain list hides
Clear priorities	Show who needs attention first
Measured action	Show what the school did and whether it helped
Parents informed	Parents receive absence alerts and a monthly summary in Khmer and English
Fast recording	A class of 40 with 2 absences takes about 4 taps

> **Do not over-claim.** Do not say "the system finds why students are absent". Attendance data alone cannot show the real reason. Say: *"the system finds attendance patterns and factors that may need further investigation."* Also avoid "our AI predicts dropouts". Use **Attendance Risk Detection / Early Warning**.

7. How teachers record attendance

Recording must be as short as possible, because good analytics needs complete, on-time data.

Step	What happens
1	Teacher logs in (Android app or website)
2	The system shows today's sessions from the timetable, for example *10B · Mathematics · Period 2 · 08:00*, with the next one highlighted
3	The system loads the enrolled students for that class. **Everyone starts as Present.**
4	Teacher taps only the absent students (with Undo). An absence reason can be added now or later (optional).
5	Teacher reviews the summary ("Present 40 · Absent 2") and submits
6	The system confirms, offers to notify the parents of absent students, and shows the next class

> **Key principle:** the teacher never chooses students manually. The system knows the class, subject and period from the teaching assignment and timetable, and loads the right list. While marking, the session is saved as a **draft**. Drafts are never counted in analytics.

8. User roles and permissions (2 roles)

The system has **two roles**. Students and parents are records in the system, not users. Parents receive messages but do not log in.

Role	Purpose	Main responsibilities
Admin	Manages the system and the school-wide analytics	Create and disable accounts; manage students, parents, teachers, classes, subjects, academic years, teaching assignments and timetables; view the school dashboard, drill-down, risk list and hotspots; manage interventions; approve correction requests; export reports; view audit logs. The principal and academic office use Admin accounts.
Teacher	Records attendance and follows up own students	Log in on the app or website; see own timetable; mark Present/Absent; add absence reasons; correct own records within 24 hours; notify parents of absent students; see history, analytics and alerts for own classes; record follow-ups for own students

Permission table

Permission	Admin	Teacher
Manage user accounts	Full access	No
Manage students and parents	Create, edit, archive	View students in assigned classes
Manage teachers	Full access	View own profile
Manage classes, subjects, timetables	Full access	View assigned classes and own timetable
Record attendance	Any session (for example a substitute teacher); always logged	Own assigned sessions only
Correct submitted attendance	Any time, with a required reason; always logged	Own sessions within 24 hours; after that, send a correction request
Approve correction requests	Yes	No
Analytics, risk list, alerts	Whole school	Assigned classes
Interventions	Create, view and close all	Create and view for own students
Parent messages	All students	Students in assigned classes
Export reports	Whole school	Assigned classes
View audit logs	Full, read-only	Own activity

> If the school later wants a separate **read-only principal account**, it can be added as a third row in the `roles` table without changing the rest of the database. It is **not** part of this version.

Attendance correction rules

- **Correction window: 24 hours.** A teacher can correct their own submitted session within 24 hours.
- **After 24 hours**, the teacher sends a **correction request** with a reason, and an Admin approves or rejects it.
- **Every change is logged**: who, when, old status, new status and reason. History is never deleted.
- The **backend** checks the time limit and permissions, not only the app screen.

Security rules

Rule	Meaning
Authentication	Users must log in before seeing any data; passwords are stored as hashes
Role-based access	Permissions depend on the role: Admin or Teacher
Assignment-based access	Teachers can only access classes and subjects assigned to them
Attendance validation	Only Present or Absent; one record per student per session
Correction history	Every correction stores the editor, time, old and new values and reason
Server-side checks	The backend checks permission on every read and edit; hiding a button is not enough

9. Early warning, interventions and parent messages NEW

Alerts

Alert	When it is created	Who sees it
Absences in a row	Absent 3 school days in a row	The student's teachers and Admin
Risk level increased	A student moves up to High or Critical	The student's teachers and Admin
Session not recorded	A session is still not submitted 30 minutes after it ends	The teacher; Admin sees the daily list

Interventions

An intervention is an action the school takes for a student. Each intervention records:

- **Type:** parent message (Telegram or SMS), phone call, meeting with parent, talk with student, counselling, other
- **Who** did it, **when**, short notes, and a **follow-up date** (default: 2 weeks later)
- **Before rate**, saved automatically when the intervention is created
- **After rate and result**, calculated on the follow-up date: *Improved*, *No clear change* or *Declined* (rules in Section 12)
- **Status:** Open → Follow-up due → Closed

> The before/after comparison shows whether attendance changed after the action. It does **not** prove that the action caused the change, so reports use the words "after the intervention", not "because of the intervention".

Parent messages

Message	When	Content
Absence alert	Teacher taps "Notify parents" after submitting	"Touch Vuthy (10B) was absent today, Period 2 (Mathematics). — Mr. Pen Sovann", in Khmer and English

Message	When	Content
Monthly report	Automatically on the last school day of each month at 17:00 (can be switched off), or sent manually	Days present out of school days, rate, and the dates absent

- **Telegram** is used when the parent has joined the school's Telegram bot; otherwise the message is sent by **SMS**.
- Parents join the bot only with consent. Every message is logged (sent, failed) and is not sent twice.
- Sending an absence alert can automatically create a "Parent message" intervention for students at High or Critical risk.

10. Screens NEW

Built = already in the current prototype (static website and Figma Make Android prototype). **To build** = planned in this version.

Website: Admin	Status
Login (Admin/Teacher), Forgot password	**Built**
Dashboard	**Built**
Take Attendance	**Built**
Attendance Monitoring	**Built**
Reports	**Built**
At-Risk Students	**Built**
Students (+ details, import)	**Built**
Classes, Subjects, Teachers (+ details)	**Built**
Parents, Teaching Assignments, Profile	**Built**
Drill-down: Grade → Class → Student Profile	**To build**
Hotspots (day × period grid)	**To build**
Interventions	**To build**
Correction Requests	**To build**
Audit Log	**To build**

Website: Teacher	Status
My Dashboard	**Built**
Take Attendance	**Built**
Attendance History	**Built**
My Classes, Profile	**Built**

Android app: Teacher	Status
Splash, Login	**Built**
Home (next class first)	**Built**
Classes, Class Details	**Built**
Mark: choose → mark → review → done	**Built**
Notify parents, Monthly report	**Built**
History, Analytics, Profile	**Built**
Absence reason picker	**To build**
Alerts and follow-up	**To build**

All screens support **English and Khmer** and **light and dark mode**, and website tables turn into cards on phones.

11. Database design (PostgreSQL)

One shared backend (Laravel REST API) and one PostgreSQL database serve both the Android app and the website. The key idea is to separate:

- **Attendance session** = one lesson on one date (for example 10B, Mathematics, Period 2, 28 Oct 2026).
- **Attendance record** = one student's Present/Absent inside that session.

Table groups

Group	Tables	Purpose
Access	roles, users, teachers	Login and permissions
School structure	academic_years, classes, subjects, students, guardians NEW, student_guardians NEW	Core school data and parent contacts
Assignments	student_enrollments, teaching_assignments, schedules	Who studies and teaches what, and when
Attendance	attendance_sessions, attendance_records, absence_reasons NEW	Recording
Analytics and action	risk_snapshots NEW, alerts NEW, interventions NEW, notifications NEW	Early warning, follow-up and parent messages
Accountability	correction_requests, audit_logs	Corrections after 24 hours and change history

Main tables

Table	What it stores	Key fields
roles	The two roles	role_id, role_name (ADMIN, TEACHER)
users	Login accounts	user_id, role_id, email, password_hash, is_active, last_login_at
teachers	Teacher info linked to a user	teacher_id, user_id, teacher_code, full_name, full_name_km, phone
academic_years	School years	academic_year_id, year_name, start_date, end_date, is_current
classes	Classes such as 10A	class_id, class_name, grade_level (10–12), shift, room, academic_year_id, homeroom_teacher_id
subjects	Mathematics, Physics, Khmer...	subject_id, subject_code, subject_name, subject_name_km
students	Student information	student_id, student_code, full_name, full_name_km, gender, date_of_birth, is_active
guardians	Parents or guardians (contacts only)	guardian_id, full_name, phone, telegram_chat_id (null if not joined), preferred_language
student_guardians	Links students to guardians	student_id, guardian_id, relationship, is_primary

Table	What it stores	Key fields
student_enrollments	Which student is in which class each year	student_id, class_id, enrolled_from, enrolled_to
teaching_assignments	Teacher + class + subject	assignment_id, teacher_id, class_id, subject_id, effective_from, effective_to
schedules	Day, period and time of each lesson	schedule_id, assignment_id, day_of_week, period_no, start_time, end_time
attendance_sessions	One lesson on one date	session_id, schedule_id, session_date, recorded_by, status (DRAFT / SUBMITTED), submitted_at
attendance_records	One student's status per session	record_id, session_id, student_id, status (PRESENT / ABSENT), reason_id (null), note, marked_by
absence_reasons	Reason list	reason_id, code (SICK, FAMILY, SCHOOL_ACTIVITY, OTHER, NOT_GIVEN), name_en, name_km
risk_snapshots	Daily risk result per student (keeps history for trends)	student_id, snapshot_date, rate_4w, rate_prev_4w, change_points, absent_streak, risk_level
alerts	Early-warning alerts	alert_id, type, student_id, class_id, session_id, created_at, status (OPEN / SEEN / CLOSED)
interventions	Actions taken for a student	intervention_id, student_id, created_by, type, action_date, notes, follow_up_date, before_rate, after_rate, outcome, status
notifications	Messages sent to guardians	notification_id, guardian_id, student_id, type (ABSENCE_ALERT / MONTHLY_REPORT), channel (TELEGRAM / SMS), status (QUEUED / SENT / FAILED), sent_at
correction_requests	Changes requested after 24 hours	request_id, record_id, old_status, requested_status, reason, requested_by, reviewed_by, decision, decided_at
audit_logs	History of important changes	log_id, user_id, action, table_name, row_id, old_values, new_values, reason, created_at

Database rules that must be enforced

Rule	Requirement
Unique attendance	Unique (session_id, student_id) in attendance_records
Unique session	Unique (schedule_id, session_date) in attendance_sessions
Valid status	status IN ('PRESENT', 'ABSENT') only
Reason only when absent	reason_id must be null when status = 'PRESENT'
Teacher authorization	The teacher must be assigned to that class and subject
Enrollment validation	The student must be enrolled in the class on the session date
Complete roster	A session can be SUBMITTED only when every enrolled student has a record

Rule	Requirement
Drafts not counted	Analytics use SUBMITTED sessions only
Correction deadline	The backend checks the 24-hour window for teachers
Audit integrity	audit_logs cannot be updated or deleted by the application account

Safety triggers

- **Guard trigger:** blocks changes to attendance unless the user is active, the session is submitted, and (for a teacher) it is their own session within 24 hours. Admin changes are allowed but always logged.
- **Audit trigger:** writes an audit_logs row whenever an attendance status or reason changes.

Triggers are an extra safety layer, not the whole security. The backend sets the logged-in user ID after authentication (never trusting an ID sent by the app), and only the backend's database account can write to the tables.

12. Analytics rules and formulas NEW

Clear, written rules make the analytics easy to explain and test. The numbers below are **starting values** that the Admin can adjust later.

Measure	Rule
Attendance rate	Present records ÷ all records in SUBMITTED sessions × 100%, for the chosen period
Trend	Rate in the last 4 school weeks minus the rate in the 4 weeks before (in percentage points). **Declining** = −5 or lower; **Improving** = +5 or higher; otherwise **Stable**.
Absence streak	Number of school days in a row on which the student was absent in all their submitted sessions
Hotspot	A day × period cell with at least 20 records whose rate is 5 or more points below the school average
Minimum data	No risk level is shown until a student has at least 10 submitted sessions in the period ("Not enough data")

Risk levels (the highest matching level is used)

Level	Rule (any one is enough)	Suggested action
Critical	Rate below 70%, or absent 5+ school days in a row	Admin review this week; contact the parent; meeting
High	Rate 70–79%, or absent 3–4 days in a row, or trend −10 points or lower	Teacher follow-up; parent message; monitor 2 weeks
Medium	Rate 80–89%, or trend −5 to −9 points, or 3+ absences with no reason given this month	Teacher keeps watch; check the reason
Low	Rate 90% or higher and trend Stable or Improving	No action

Intervention result

- **Before rate** = rate in the 10 school days before the action; **after rate** = rate in the 10 school days after it.

- **Improved** = +5 points or more; **Declined** = −5 points or less; otherwise **No clear change**.
- If either side has fewer than 5 sessions, the result is "Not enough data".

Missing records

A rate can look too high if some records are missing. The backend therefore refuses to submit a session until every enrolled student has a Present or Absent record (rule "Complete roster" in Section 11).

13. Scope, timeline and development order NEW

Scope

Priority	Features
Must have (first release)	Login with 2 roles; school data management; timetables; attendance recording on app and website; drafts and submit; 24-hour correction and audit log; descriptive analytics (rate by student, class, grade, subject, month); trend; risk levels and risk list; absences-in-a-row alert; Student Profile; English and Khmer
Should have	Absence reasons; hotspots; drill-down; interventions with before/after result; parent absence alerts and monthly report (Telegram, SMS fallback); correction requests; Excel/PDF export
Future	Offline marking with sync; QR check-in; push notifications; read-only principal role; prediction models (only with enough historical data)
Not included	Late, Excused or Leave as statuses (status is Present or Absent only); student or parent logins

Suggested timeline (12 weeks; adjust to the semester)

Weeks	Work	Result
1–2	Final requirements; database migrations and seed data	Tested database
3–4	Backend: login, 2 roles, permissions, school data, timetables	Working API for admin data
5–6	Attendance sessions, submit, 24-hour correction, audit log; connect Android app	Real attendance recording
7–8	Analytics queries: rate, trend, risk, streaks; dashboards	Live dashboards
9–10	Alerts, interventions, parent messages, reasons, hotspots, drill-down	Early-warning loop
11–12	Testing with realistic data; fixes; report and presentation	Final demo

Development order

1. Finalize and test the database migrations (the design work comes first, to avoid rework).
2. Authentication and role/assignment permissions in the shared backend.
3. Enrollments, teaching assignments and timetables.
4. Attendance sessions, submission and correction.
5. Analytics queries, risk snapshots and reports.
6. Alerts, interventions and parent messages.
7. Connect the Android app and the website to the same API.

8. Test security and calculations with realistic school data.

14. Testing, privacy and limitations NEW

Acceptance tests (examples)

Test	Expected result
Teacher opens a class not assigned to them	Blocked by the backend (403), even if called directly
Two devices submit the same session at the same time	One session and one record per student; no duplicates
Teacher edits a record after 24 hours	Blocked; a correction request can be sent
A record is corrected	The audit log shows old and new values, user, time and reason
A session is left as a draft	Not counted in any rate
A student is absent 3 school days in a row	An alert is created once for their teachers and Admin
Rate 76% and trend −3	Risk level = High
Parent not on Telegram	The message is sent by SMS and logged
Class of 40, 2 absent	Recorded and submitted in about 4 taps, in under 1 minute

Privacy and ethics

- Attendance is personal data about minors. Only Admins and assigned teachers can see it; all access is through login.
- Parents join Telegram messages with consent and can stop them; phone numbers are not shown to other parents.
- Analytics describe patterns, not people's character. Teachers and students are never ranked or blamed in reports.
- Risk levels are a reason to talk to a student, not a punishment.
- Old records are kept for the academic year plus one year, then archived according to school policy.

Known limitations

- Attendance data shows **patterns**, not the real reasons behind them.
- Before/after results show change, not proof of cause.
- Risk rules are simple and transparent on purpose; they should be reviewed after one term of real data.
- The current screens are a prototype with sample data; they become live after the backend is connected.

15. Appendix: what changed from v1.0

Topic	v1.0	v2.0
Roles	3 roles (Admin, School Director, Teacher)	**2 roles** (Admin, Teacher); the principal uses an Admin account

Topic	v1.0	v2.0
Who uses the website	Admins and director	Admins, and teachers for their own classes
Late and reasons	Mentioned late counts and reasons, but only Present/Absent allowed	Status is Present/Absent only; optional absence reason; no "late"
Grades in examples	Grade 7, 8, 9	Grade 10, 11, 12 (high school)
Risk level	Described, no exact rules	Exact rules for Critical, High, Medium and Low, plus minimum data
Trend and hotspot	Described	Exact formulas (4 weeks vs 4 weeks; 5-point threshold)
Interventions	Main idea, but no table	interventions table, types, follow-up date, before/after rule
Parents	Not in the database	guardians and notifications tables; Telegram with SMS fallback; monthly report
Alerts	Described	alerts table and three alert types
Screens	Not listed	Full screen list with Built / To build status
Scope and timeline	Development order only	Must / Should / Future list and a 12-week timeline
Testing and privacy	Checklist only	Acceptance tests, privacy rules and limitations